

TED (10)4040

Reg. No.

(REVISION 2010)

Signature.

FIFTH SEMESTER DIPLOMA EXAMINATION IN ELECTRICAL AND ELECTRONICS
ENGINEERING

MODEL EXAMINATION

[Time: 3 Hours

POWER SYSTEM PROTECTION

Marks

PART-A

- I. Answer all questions in one or two sentences. Each question carries 2 marks.
1. Define the term fusing factor.
 2. Define the term re-striking voltage.
 3. What is the necessity of Back Up protection?
 4. List out any two abnormal operating conditions that are likely to occur in an alternator.
 5. Give any two methods of neutral earthing. [2x5=10

PART- B

- II Answer any **five** questions. Each question carries 6 marks
1. Classify fuses.
 2. Explain I^2Rt characteristics of HRC fuse with graph.
 - 3 Write any six advantages of SF6 circuit breaker.
 - 4 What are the duties of a circuit breaker?
 - 5 Explain the working of an attraction armature type relay with the aid of figure.
 6. Explain the procedure for determining the operating time of an induction relay.
 7. List out any six tests carried out on a lightning arrester. [6x5=30

PART – C

(Answer one question from each unit. Each question carries 15 marks)

UNIT -I

- III a) Write the advantages and disadvantages of rewirable fuse. 8
- b) With neat graph correlate cut off characteristics of HRC fuse. 7

OR

- c) What are the minimum requirements for a material to be used as fuse material? 7
- d) Give the advantages and disadvantages of HRC fuse. 8

UNIT-II

- IV a) With neat sketch explain the construction and working of plain break oil circuit breaker. 8
- b) Discuss the problems associated with SF6 circuit breaker. 7

OR

- c) With neat sketch explain the construction and working of vacuum circuit breaker. 8
- d) Compare Air blast circuit breakers and SF6 circuit breakers. 7

UNIT-III

- V a) List out the important properties of a good protective relay system. 8
- b) Explain primary and back up protection in a power system 7

OR

- VI a) Explain the working of an attracted armature type relay with a neat diagram. 8
- b) Classify protective relays according to their principles of operation. 7

UNIT – IV

- VII a) Draw and explain the schematic arrangement of Merz –price protection provided for a three phase alternator 8
- b) Explain the working of an expulsion type lightning arrester. Also state its drawbacks. 7

OR

- VII a) Explain the working of a valve type lightning arrester. Also draw its volt-ampere characteristic. 8
- b) Explain the resonant neutral grounding method with the aid of diagram. 7

[15x 4=60]